

## 10.10 Convergent and Power-Bounded Matrices

Chapter 9 asked what happens to  $\mathbf{A}^m$  as  $m$  grows, and could answer only for diagonalizable matrices: the powers of  $\mathbf{A}$  behave like the powers of its eigenvalues. For a Markov chain that left a gap, because a transition matrix need not be diagonalizable. The Jordan form closes the gap completely. This section characterizes, in terms of the eigenvalues and the block sizes alone, when  $\mathbf{A}^m$  tends to zero, when it stays bounded, when it converges, and when the geometric series  $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \cdots$  converges.

Convergence here means what it meant in Chapter 9: **entrywise**. No notion of the size of a matrix is used. Chapter 16 introduces norms and re-proves several of these results in a way that also gives numerical bounds; the price of doing without them is that we must look at one entry at a time, and the Jordan form is what makes that possible.

### 10.10.1 Entrywise limits and bounds

#### Definition 10.10.1. Entrywise Convergence and Boundedness

Let  $\mathbf{M}_1, \mathbf{M}_2, \dots$  be matrices in  $M_{p \times q}(\mathbb{C})$  and  $\mathbf{M} \in M_{p \times q}(\mathbb{C})$ .

- (a) The sequence  $(\mathbf{M}_m)$  **converges** to  $\mathbf{M}$ , written  $\mathbf{M}_m \rightarrow \mathbf{M}$ , if for **every** position  $(r, s)$  the sequence of complex numbers  $(\mathbf{M}_m)_{rs}$  converges to  $\mathbf{M}_{rs}$ . We then write  $\mathbf{M} = \lim_m \mathbf{M}_m$ .
- (b) The sequence  $(\mathbf{M}_m)$  is **bounded** if there is a real  $c \geq 0$  with  $|(\mathbf{M}_m)_{rs}| \leq c$  for **all**  $m$  and **all** positions  $(r, s)$ .
- (c) A series  $\sum_{k \geq 0} \mathbf{M}_k$  **converges** to  $\mathbf{M}$  if its sequence of partial sums  $\sum_{k=0}^K \mathbf{M}_k$  converges to  $\mathbf{M}$ .

In words: nothing new is happening. Each of the  $pq$  positions is watched separately, and the words mean for matrices exactly what they mean for the finitely many number sequences at those positions. Since a finite set of bounds can be replaced by their maximum, (b) is the same as asking each entry sequence to be bounded.

*Remark. What we assume from analysis.* Beyond the four facts quoted in Chapter 9 §11, this section uses exactly the following.

- (B1) **Polynomial against geometric decay.** For every integer  $j \geq 0$  and every real  $r$  with  $0 \leq r < 1$ ,  $m^j r^m \rightarrow 0$  as  $m \rightarrow \infty$ .
- (B2) **Growth.** For real  $r > 1$ , the sequence  $(r^m)$  is unbounded.
- (B3) **Limits.** Limits of convergent sequences of complex numbers respect sums, scalar multiples and products; a convergent sequence is bounded; every subsequence of a convergent sequence converges to the same limit; and  $z_m \rightarrow z$  implies  $|z_m| \rightarrow |z|$ .
- (B4) **Terms of a convergent series.** If  $\sum_k z_k$  converges, then  $z_k \rightarrow 0$ .

Two reductions make every proof below short. The first moves the question along a similarity, so that we may assume  $\mathbf{A}$  is a Jordan matrix. The second computes the powers of a single Jordan block.

**Lemma 10.10.1. Convergence and Boundedness Survive a Similarity**

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  with  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  for an invertible  $\mathbf{P} \in M_n(\mathbb{C})$ , and let  $\mathbf{L} \in M_n(\mathbb{C})$ . Then:

- (a)  $\mathbf{A}^m \rightarrow \mathbf{L}$  if and only if  $\mathbf{B}^m \rightarrow \mathbf{P}^{-1}\mathbf{L}\mathbf{P}$ ;
- (b)  $(\mathbf{A}^m)$  is bounded if and only if  $(\mathbf{B}^m)$  is bounded.

*Proof.* By Proposition 4.2.6 (c),  $\mathbf{B}^m = \mathbf{P}^{-1}\mathbf{A}^m\mathbf{P}$  for every  $m \geq 0$ . Writing out the products (Definition 0.9.9), the  $(r, s)$ -entry of  $\mathbf{B}^m$  is

$$(\mathbf{B}^m)_{rs} = \sum_{u,v} (\mathbf{P}^{-1})_{ru} (\mathbf{A}^m)_{uv} \mathbf{P}_{vs},$$

a linear combination of the entries of  $\mathbf{A}^m$  with coefficients that do not depend on  $m$ . So if  $\mathbf{A}^m \rightarrow \mathbf{L}$ , then by (B3) each entry of  $\mathbf{B}^m$  converges to the matching combination of the entries of  $\mathbf{L}$ , that is,  $\mathbf{B}^m \rightarrow \mathbf{P}^{-1}\mathbf{L}\mathbf{P}$ ; and if  $|(\mathbf{A}^m)_{uv}| \leq c$  for all  $m, u, v$ , then  $|(\mathbf{B}^m)_{rs}| \leq c \sum_{u,v} |(\mathbf{P}^{-1})_{ru}| |\mathbf{P}_{vs}|$ , a bound independent of  $m$ , so  $(\mathbf{B}^m)$  is bounded. The converses follow by exchanging the roles of  $\mathbf{A}$  and  $\mathbf{B}$ , using  $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ . ■

For the second reduction, recall from Lemma 10.9.1, applied to the polynomial  $x^m$ , that

$$(\mathbf{J}_k(\lambda)^m)_{r,r+j} = \binom{m}{j} \lambda^{m-j} \quad (0 \leq j \leq k-1),$$

with zeros below the diagonal. Everything in this section is read off this one formula. Two remarks about it: the binomial coefficient is bounded by  $m^j$ , since

$$\binom{m}{j} = \frac{m(m-1) \cdots (m-j+1)}{j!} \leq \frac{m^j}{j!} \leq m^j \quad (m \geq j \geq 1),$$

and  $\binom{m}{1} = m$  grows without bound, while  $\binom{m}{0} = 1$  is constant. So the only question is whether the geometric factor  $\lambda^m$  beats the polynomial factor  $m^j$ .

*Quick check.* Compute  $|(\mathbf{J}_3(\lambda)^m)_{1,3}|$  for  $\lambda = \frac{1}{2}$  and for  $\lambda = 1$ , and say in each case whether it tends to 0, stays bounded, or grows without bound.

*Solution.* Here  $j = 2$ , so the entry is  $\binom{m}{2} \lambda^{m-2}$ , of modulus  $\frac{m(m-1)}{2} |\lambda|^{m-2}$ . For  $\lambda = \frac{1}{2}$  this is  $2 \cdot \frac{m(m-1)}{2} 2^{-m} \leq 2m^2 2^{-m}$ , which tends to 0 by (B1) with  $j = 2$  and  $r = \frac{1}{2}$ . For  $\lambda = 1$  it is  $\frac{m(m-1)}{2}$ , which grows without bound. The modulus of  $\lambda$  decides, and when  $|\lambda| = 1$  the block size decides.

**10.10.2 Powers tending to zero**

*The powers of  $\mathbf{A}$  die out exactly when every eigenvalue has modulus less than one.*

**Theorem 10.10.2. Powers Tending to Zero**

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ . Then  $\mathbf{A}^m \rightarrow 0$  if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| < 1$ .

*Idea.* ( $\Leftarrow$ ) Pass to the Jordan form and look at one entry: it is  $\binom{m}{j}\lambda^{m-j}$ , a polynomial in  $m$  times a geometric sequence that decays, and geometric decay wins. ( $\Rightarrow$ ) The other direction needs no Jordan form at all: an eigenvector turns  $\mathbf{A}^m$  into  $\lambda^m$ .

*Proof.* ( $\Leftarrow$ ) Suppose  $|\lambda| < 1$  for every  $\lambda \in \text{spec}(\mathbf{A})$ . By Theorem 10.3.1,  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  a Jordan matrix, whose diagonal entries  $\lambda_i$  are the eigenvalues of  $\mathbf{A}$  (Corollary 10.3.2 (a)). By Theorem 8.1.4 (b),  $\mathbf{J}^m$  is the direct sum of the blocks  $\mathbf{J}_{k_i}(\lambda_i)^m$ , so each entry of  $\mathbf{J}^m$  is either 0 for every  $m$  or of the form  $\binom{m}{j}\lambda_i^{m-j}$  with  $0 \leq j \leq k_i - 1$ . For the latter, with  $r = |\lambda_i| < 1$ , if  $r = 0$  the entry is 0 for  $m > j$ ; otherwise, for  $m \geq j$ ,

$$\left| \binom{m}{j} \lambda_i^{m-j} \right| \leq m^j r^{m-j} = r^{-j} m^j r^m \rightarrow 0$$

by (B1), the factor  $r^{-j}$  being a constant. So  $\mathbf{J}^m \rightarrow 0$ , and  $\mathbf{A}^m \rightarrow \mathbf{P}0\mathbf{P}^{-1} = 0$  by Lemma 10.10.1 (a).

( $\Rightarrow$ ) Suppose  $\mathbf{A}^m \rightarrow 0$  and let  $\lambda \in \text{spec}(\mathbf{A})$  with eigenvector  $\mathbf{v} \neq \mathbf{0}$ . Then  $\mathbf{A}^m \mathbf{v} = \lambda^m \mathbf{v}$  for every  $m$  (induction on  $m$ ). Pick a coordinate  $r$  with  $v_r \neq 0$ . The  $r$ -th entry of  $\mathbf{A}^m \mathbf{v}$  is a linear combination of entries of  $\mathbf{A}^m$  with coefficients  $v_1, \dots, v_n$ , so it tends to 0 by (B3); that entry is  $\lambda^m v_r$ , so  $\lambda^m \rightarrow 0$  and hence  $|\lambda|^m = |\lambda^m| \rightarrow 0$  by (B3). If  $|\lambda| \geq 1$ , then  $|\lambda|^m \geq 1$  for every  $m$  and the limit cannot be 0. Hence  $|\lambda| < 1$ . This proves the theorem. ■

Note the asymmetry: the hard direction needs the Jordan form, the easy one needs only an eigenvector. That pattern repeats in the next two theorems.

*Warning.* **The condition is on the eigenvalues, not on the entries.** Every entry of  $\mathbf{A} = \begin{pmatrix} 0.9 & 0.9 \\ 0.9 & 0.9 \end{pmatrix}$  has modulus less than 1, yet  $\mathbf{A}\mathbf{1} = 1.8\mathbf{1}$ , so 1.8 is an eigenvalue and  $\mathbf{A}^m \mathbf{1} = 1.8^m \mathbf{1}$  grows without bound. Conversely, a matrix with huge entries can have all eigenvalues 0: the matrix  $\begin{pmatrix} 0 & 10^6 \\ 0 & 0 \end{pmatrix}$  squares to 0. Nothing can be concluded from the size of the entries alone.

**10.10.3 Powers that stay bounded**

Dropping “tends to zero” to “stays bounded” allows eigenvalues on the unit circle, but only if they carry no chains: a block of size  $\geq 2$  contributes an entry  $\binom{m}{1}\lambda^{m-1}$  of modulus  $m$ .

**Theorem 10.10.3. Power-Bounded Matrices**

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with Jordan form  $\mathbf{J}$ . The sequence  $(\mathbf{A}^m)_{m \geq 0}$  is bounded if and only if

- (i) every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| \leq 1$ , and
- (ii) every eigenvalue  $\lambda$  with  $|\lambda| = 1$  has **all** of its blocks in  $\mathbf{J}$  of size 1, equivalently  $a_{\mathbf{A}}(\lambda) = g_{\mathbf{A}}(\lambda)$ , equivalently  $x - \lambda$  occurs to the first power in  $m_{\mathbf{A}}$ .

*Proof.* Write  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  (Theorem 10.3.1). By Lemma 10.10.1 (b),  $(\mathbf{A}^m)$  is bounded if and only if  $(\mathbf{J}^m)$  is, and by Theorem 8.1.4 (b) the latter holds if and only if every block sequence  $(\mathbf{J}_{k_i}(\lambda_i)^m)$  is bounded. The three reformulations in (ii) are Corollary 10.3.2 (a), (b) and (c): all  $\lambda$ -blocks have size 1 exactly when their number equals their total size, and exactly when the largest of their sizes is 1.

( $\Leftarrow$ ) Assume (i) and (ii), and fix a block  $\mathbf{J}_k(\lambda)$  of  $\mathbf{J}$ . If  $|\lambda| < 1$ , then  $\mathbf{J}_k(\lambda)^m \rightarrow 0$  by Theorem 10.10.2 applied to the single block  $\mathbf{J}_k(\lambda)$ , whose only eigenvalue is  $\lambda$ , hence  $(\mathbf{J}_k(\lambda)^m)$  is bounded by (B3). If  $|\lambda| = 1$ , then  $k = 1$  by (ii), so  $\mathbf{J}_k(\lambda)^m = (\lambda^m)$  is a  $1 \times 1$  matrix of modulus  $|\lambda|^m = 1$ . Either way the block sequence is bounded, so  $(\mathbf{J}^m)$  is bounded and so is  $(\mathbf{A}^m)$ .

( $\Rightarrow$ ) Assume  $(\mathbf{A}^m)$  is bounded, hence so is  $(\mathbf{J}^m)$ , hence so is every block sequence.

*Proof of (i).* Let  $\lambda$  be an eigenvalue and  $\mathbf{J}_k(\lambda)$  one of its blocks. The  $(1, 1)$ -entry of  $\mathbf{J}_k(\lambda)^m$  is  $\lambda^m$ , so  $(|\lambda|^m)$  is bounded; by (B2) this forces  $|\lambda| \leq 1$ .

*Proof of (ii).* Suppose some eigenvalue  $\lambda$  with  $|\lambda| = 1$  had a block  $\mathbf{J}_k(\lambda)$  with  $k \geq 2$ . Its  $(1, 2)$ -entry at the  $m$ -th power is  $\binom{m}{1}\lambda^{m-1} = m\lambda^{m-1}$ , of modulus  $m|\lambda|^{m-1} = m$ , which is unbounded — contradicting the boundedness of that block sequence. So every such block has size 1. This proves the theorem. ■

## 10.10.4 Powers that converge

Boundedness is not convergence:  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^m$  alternates between two matrices forever. Among the eigenvalues of modulus 1, only  $\lambda = 1$  has convergent powers, because  $\lambda^m$  has to sit still.

### Theorem 10.10.4. Convergent Matrices

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with Jordan form  $\mathbf{J}$ . Then  $(\mathbf{A}^m)$  converges if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies

- (i)  $|\lambda| < 1$ , or
- (ii)  $\lambda = 1$  and all of its blocks in  $\mathbf{J}$  have size 1.

In that case

$$\lim_{m \rightarrow \infty} \mathbf{A}^m = \mathbf{\Pi},$$

where  $\mathbf{\Pi} \in M_n(\mathbb{C})$  is the matrix, in the standard basis, of the projection onto  $E_1(\mathbf{A})$  along  $\bigoplus_{\lambda \neq 1} G_{\lambda}(\mathbf{A})$  if  $1 \in \text{spec}(\mathbf{A})$ , and  $\mathbf{\Pi} = 0$  otherwise.

*Idea.* For the shape of the answer, take the Jordan form and let  $m \rightarrow \infty$  in each block: the blocks with  $|\lambda| < 1$  go to 0, the blocks  $\mathbf{J}_1(1) = (1)$  stay at 1. So  $\mathbf{J}^m$  tends to the 0/1 diagonal matrix that marks the 1-blocks, which is the matrix of a projection; conjugating back identifies the two subspaces. For the converse, the  $(1, 1)$ -entry of a block is  $\lambda^m$ , and if that converges to  $z$  then so does  $\lambda^{m+1} = \lambda \cdot \lambda^m$ , giving  $z = \lambda z$  and forcing  $\lambda = 1$  or  $z = 0$ .

*Proof.* Write  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  (Theorem 10.3.1), and let  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  be the corresponding Jordan basis, the columns of  $\mathbf{P}$ . For each  $i$  let  $C_i \subseteq \mathbb{C}^n$  be the span of the  $k_i$  basis vectors belonging to block  $i$ , so that  $\mathbb{C}^n = C_1 \oplus \cdots \oplus C_d$ .

( $\Leftarrow$ ) Suppose every eigenvalue satisfies (i) or (ii). Fix a block. If  $|\lambda_i| < 1$ , then  $\mathbf{J}_{k_i}(\lambda_i)^m \rightarrow 0$  by Theorem 10.10.2 applied to the block  $\mathbf{J}_{k_i}(\lambda_i)$ , whose only eigenvalue is  $\lambda_i$ . If  $\lambda_i = 1$ , then  $k_i = 1$  by (ii) and  $\mathbf{J}_1(1)^m = (1)$  for every  $m$ . By Theorem 8.1.4 (b),  $\mathbf{J}^m$  is the direct sum of these, so

$$\mathbf{J}^m \rightarrow \mathbf{E},$$

where  $\mathbf{E}$  is the diagonal matrix carrying 1 in the positions of the blocks with  $\lambda_i = 1$  and 0 everywhere else. By Lemma 10.10.1 (a),  $\mathbf{A}^m \rightarrow \mathbf{PEP}^{-1}$ .

It remains to identify  $\mathbf{PEP}^{-1}$ . The matrix  $\mathbf{E}$  is the matrix in the basis  $\mathcal{B}$  of the projection onto

$$U := \bigoplus_{\lambda_i=1} C_i \quad \text{along} \quad W := \bigoplus_{\lambda_i \neq 1} C_i,$$

because  $\mathbf{E}$  fixes each basis vector of the first family and kills each of the second (Theorem 4.5.1). Since the columns of  $\mathbf{P}$  are the vectors of  $\mathcal{B}$ ,  $\mathbf{P}$  is the change-of-coordinates matrix from  $\mathcal{B}$  to the standard basis, so  $\mathbf{PEP}^{-1}$  is the matrix of that projection in the standard basis (Theorem 4.2.3). Now, for any  $\mu \in \text{spec}(\mathbf{A})$ ,

$$\bigoplus_{\lambda_i=\mu} C_i = G_\mu(\mathbf{A}).$$

Indeed each  $C_i$  with  $\lambda_i = \mu$  is killed by  $(\mathbf{A} - \mu\mathbf{I})^{k_i}$ , hence lies in  $G_\mu(\mathbf{A})$  (Corollary 10.1.2 (a)); the sum on the left is direct and has dimension  $\sum_{\lambda_i=\mu} k_i = a_{\mathbf{A}}(\mu)$  by Corollary 10.3.2 (a); and  $\dim G_\mu(\mathbf{A}) = a_{\mathbf{A}}(\mu)$  by Theorem 10.1.3 (c). An inclusion of subspaces with equal dimensions is an equality (Theorem 1.6.8, applied inside  $G_\mu(\mathbf{A})$ ). Hence  $W = \bigoplus_{\lambda \neq 1} G_\lambda(\mathbf{A})$ , and  $U = G_1(\mathbf{A})$  if  $1 \in \text{spec}(\mathbf{A})$ , with  $U = \{0\}$  otherwise. Finally, when  $1 \in \text{spec}(\mathbf{A})$ , condition (ii) says all 1-blocks have size 1, so  $g_{\mathbf{A}}(1) = a_{\mathbf{A}}(1)$  by Corollary 10.3.2 (a), (b); since  $E_1(\mathbf{A}) \subseteq G_1(\mathbf{A})$  with  $\dim E_1(\mathbf{A}) = g_{\mathbf{A}}(1) = a_{\mathbf{A}}(1) = \dim G_1(\mathbf{A})$ , we get  $G_1(\mathbf{A}) = E_1(\mathbf{A})$ . So  $\lim_m \mathbf{A}^m = \mathbf{\Pi}$  as stated.

( $\Rightarrow$ ) Suppose  $(\mathbf{A}^m)$  converges. By Lemma 10.10.1 (a) so does  $(\mathbf{J}^m)$ , hence so does each block sequence. Fix a block  $\mathbf{J}_k(\lambda)$ .

*Case  $|\lambda| > 1$ .* Impossible: a convergent sequence is bounded (B3), so this is excluded by Theorem 10.10.3.

*Case  $|\lambda| = 1$  and  $\lambda \neq 1$ .* The  $(1, 1)$ -entries give a convergent sequence  $\lambda^m \rightarrow z$ . The sequence  $(\lambda^{m+1})_{m \geq 0}$  is a tail of the same sequence, so it also converges to  $z$ ; on the other hand  $\lambda^{m+1} = \lambda \cdot \lambda^m \rightarrow \lambda z$  by (B3). Hence  $z = \lambda z$ , that is,  $(1 - \lambda)z = 0$ , and  $\lambda \neq 1$  forces  $z = 0$ . But  $|\lambda^m| = 1$  for every  $m$ , so  $|z| = 1$  by (B3), a contradiction. So this case does not occur either.

*Case  $\lambda = 1$ .* If  $k \geq 2$ , the  $(1, 2)$ -entry of  $\mathbf{J}_k(1)^m$  is  $\binom{m}{1}1^{m-1} = m$ , which does not converge. So  $k = 1$ .

The three cases leave exactly (i) and (ii). This proves the theorem. ■

The limit deserves a sentence. It is a projection, so  $\mathbf{\Pi}^2 = \mathbf{\Pi}$  — which had to be true, since  $\mathbf{\Pi}^2 = \lim_m \mathbf{A}^m \lim_m \mathbf{A}^m = \lim_m \mathbf{A}^{2m} = \mathbf{\Pi}$  by (B3) — and it acts as the identity on the eigenvectors for 1 and annihilates everything belonging to the other eigenvalues. In the long run,  $\mathbf{A}$  forgets all of  $\mathbb{C}^n$  except the fixed space  $E_1(\mathbf{A})$ .

### Example. Deciding and Computing Limits of Powers

For each matrix, decide whether  $\mathbf{A}^m$  tends to 0, stays bounded, converges, and compute  $\lim \mathbf{A}^m$  when it exists.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 1 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}_1} = x(x-2)+1 = (x-1)^2$ , so the only eigenvalue is 1. Put  $\mathbf{N} = \mathbf{A}_1 - \mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ ;

then  $\mathbf{N} \neq 0$  and  $\mathbf{N}^2 = \begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$ , so  $g_{\mathbf{A}_1}(1) = 2 - \text{rank } \mathbf{N} = 1$  and the Jordan form is  $\mathbf{J}_2(1)$ . Condition (ii) of [Theorem 10.10.3](#) fails, so  $(\mathbf{A}_1^m)$  is **unbounded**, and in particular does not converge. Explicitly,  $\mathbf{I}$  and  $\mathbf{N}$  commute and  $\mathbf{N}^2 = 0$ , so the binomial theorem gives

$$\mathbf{A}_1^m = (\mathbf{I} + \mathbf{N})^m = \mathbf{I} + m\mathbf{N} = \begin{pmatrix} 1+m & m \\ -m & 1-m \end{pmatrix},$$

which grows linearly.

- (b)  $p_{\mathbf{A}_2} = x^2 + \frac{1}{4}$ , with roots  $\pm \frac{i}{2}$ , both of modulus  $\frac{1}{2} < 1$ . By [Theorem 10.10.2](#),  $\mathbf{A}_2^m \rightarrow 0$  (so also bounded and convergent). Explicitly  $\mathbf{A}_2^2 = -\frac{1}{4}\mathbf{I}$ , so  $\mathbf{A}_2^{2l} = (-\frac{1}{4})^l \mathbf{I}$  and  $\mathbf{A}_2^{2l+1} = (-\frac{1}{4})^l \mathbf{A}_2$ , and every entry has modulus at most  $4^{-l} \rightarrow 0$ .
- (c)  $\mathbf{A}_3 = (1) \oplus \mathbf{J}_2(\frac{1}{3})$  is already a Jordan matrix. Its eigenvalues are 1, with a single block of size 1, and  $\frac{1}{3}$ , with  $|\frac{1}{3}| < 1$ . So condition (i) or (ii) holds for each eigenvalue and  $\mathbf{A}_3^m$  converges by [Theorem 10.10.4](#). With  $\mathbf{P} = \mathbf{I}$ , the matrix  $\mathbf{E}$  of the proof is  $\text{diag}(1, 0, 0)$ , so

$$\lim_m \mathbf{A}_3^m = \text{diag}(1, 0, 0),$$

the projection onto  $E_1(\mathbf{A}_3) = \text{span}(\mathbf{e}_1)$  along  $G_{1/3}(\mathbf{A}_3) = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$ . Explicitly, by

Lemma 10.9.1,

$$\mathbf{A}_3^m = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3^{-m} & m 3^{-(m-1)} \\ 0 & 0 & 3^{-m} \end{pmatrix},$$

and  $m 3^{-(m-1)} = 3m 3^{-m} \rightarrow 0$  by (B1).

### 10.10.5 Markov chains, finished

Chapter 9 proved that a Markov chain settles down under three hypotheses:  $\mathbf{A}$  diagonalizable over  $\mathbb{C}$ ,  $a_{\mathbf{A}}(1) = 1$ , and  $|\lambda| < 1$  for every other eigenvalue (Theorem 9.11.4). The first hypothesis can now be dropped, and more: for a stochastic matrix, the block condition of Theorem 10.10.3 holds automatically, because the powers of a stochastic matrix have entries in  $[0, 1]$ .

#### Corollary 10.10.5. Convergence of a Markov Chain

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic (Definition 9.11.1),  $n \geq 1$ . Then:

- (a)  $(\mathbf{A}^m)$  is bounded; consequently every eigenvalue  $\lambda \in \mathbb{C}$  of  $\mathbf{A}$  satisfies  $|\lambda| \leq 1$ , and every eigenvalue with  $|\lambda| = 1$  has only  $1 \times 1$  blocks. In particular  $a_{\mathbf{A}}(1) = g_{\mathbf{A}}(1)$ .
- (b)  $(\mathbf{A}^m)$  converges if and only if 1 is the **only** eigenvalue of  $\mathbf{A}$  of modulus 1; and then  $\lim_m \mathbf{A}^m$  is the projection onto  $E_1(\mathbf{A})$  along  $\bigoplus_{\lambda \neq 1} G_{\lambda}(\mathbf{A})$ .

*Proof.*

- (a) By Proposition 9.11.3 (b), every  $\mathbf{A}^m$  is stochastic, so its entries are non-negative and each of its columns sums to 1; hence every entry lies in  $[0, 1]$  and  $(\mathbf{A}^m)$  is bounded with  $c = 1$ . Now apply the  $(\Rightarrow)$  direction of Theorem 10.10.3. Since  $1 \in \text{spec}(\mathbf{A})$  by Proposition 9.11.3 (c), the 1-blocks exist and all have size 1, so  $a_{\mathbf{A}}(1) = g_{\mathbf{A}}(1)$  by Corollary 10.3.2 (a), (b).
- (b) By (a), every eigenvalue has  $|\lambda| \leq 1$  and every eigenvalue of modulus 1 has only  $1 \times 1$  blocks. So the criterion of Theorem 10.10.4 — each eigenvalue has  $|\lambda| < 1$ , or equals 1 with only  $1 \times 1$  blocks — holds precisely when no eigenvalue  $\lambda \neq 1$  has  $|\lambda| = 1$ . The limit is the one given by Theorem 10.10.4. ■

So the only obstruction to a Markov chain settling down is an eigenvalue of modulus 1 other than 1 itself — periodicity, in the language of probability. Chapter 9's warning example, the swap  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , has the eigenvalue  $-1$  and is exactly this obstruction.

#### Example. A Two-State Chain

Let  $\mathbf{A} = \begin{pmatrix} 3/4 & 1/2 \\ 1/4 & 1/2 \end{pmatrix}$ . Decide whether  $\mathbf{A}^m$  converges and find the limit.

*Solution.*  $\mathbf{A}$  is stochastic: the entries are non-negative, and each column sums to 1. Its characteristic polynomial is

$$p_{\mathbf{A}} = (x - \frac{3}{4})(x - \frac{1}{2}) - \frac{1}{8} = x^2 - \frac{5}{4}x + \frac{1}{4} = (x - 1)(x - \frac{1}{4}),$$

so  $\text{spec}(\mathbf{A}) = \{1, \frac{1}{4}\}$  and the only eigenvalue of modulus 1 is 1. By [Corollary 10.10.5](#) (b),  $\mathbf{A}^m$  converges.

For the limit,  $E_1(\mathbf{A}) = \ker(\mathbf{A} - \mathbf{I})$ , and  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/4 & 1/2 \\ 1/4 & -1/2 \end{pmatrix}$  gives  $\text{span}((2, 1))$ ; the steady state is  $\mathbf{v} = (\frac{2}{3}, \frac{1}{3})$ , the multiple of  $(2, 1)$  whose entries sum to 1. Also  $E_{1/4}(\mathbf{A}) = \ker(\mathbf{A} - \frac{1}{4}\mathbf{I})$ , and  $\mathbf{A} - \frac{1}{4}\mathbf{I} = \begin{pmatrix} 1/2 & 1/2 \\ 1/4 & 1/4 \end{pmatrix}$  gives  $\text{span}((1, -1))$ . Since the two eigenvalues are distinct,  $G_{1/4}(\mathbf{A}) = E_{1/4}(\mathbf{A})$  and the limit  $\mathbf{\Pi}$  is the projection onto  $\text{span}((2, 1))$  along  $\text{span}((1, -1))$ . Computing it in the basis  $\mathcal{C} = ((2, 1), (1, -1))$ : with  $\mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ ,  $\det \mathbf{Q} = -3$  and  $\mathbf{Q}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$ , so

$$\mathbf{\Pi} = \mathbf{Q} \text{diag}(1, 0) \mathbf{Q}^{-1} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & 1/3 \end{pmatrix}.$$

*Check.*  $\mathbf{\Pi} = \mathbf{v}\mathbf{1}^\top$ , as Theorem 9.11.4 predicts;  $\mathbf{\Pi}^2 = \mathbf{\Pi}$  since  $\mathbf{1}^\top \mathbf{v} = 1$ ; the columns of  $\mathbf{\Pi}$  are the steady state, so every initial probability vector  $\mathbf{x}_0$  gives  $\mathbf{x}_m = \mathbf{A}^m \mathbf{x}_0 \rightarrow \mathbf{\Pi} \mathbf{x}_0 = \mathbf{v}(\mathbf{1}^\top \mathbf{x}_0) = \mathbf{v}$ . And  $\mathbf{A}^2 = \begin{pmatrix} 11/16 & 5/8 \\ 5/16 & 3/8 \end{pmatrix}$  is already close to  $\mathbf{\Pi}$ , the error decaying like  $4^{-m}$ .

## 10.10.6 The geometric series of a matrix

The scalar identity  $\sum_k a^k = (1 - a)^{-1}$  for  $|a| < 1$  has a matrix version, with the same proof and the same hypothesis read off the eigenvalues.

### Theorem 10.10.6. Neumann Series

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ . Then the series  $\sum_{k \geq 0} \mathbf{A}^k$  converges if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| < 1$ . In that case  $\mathbf{I} - \mathbf{A}$  is invertible and

$$\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I} - \mathbf{A})^{-1}.$$

*Idea.* Telescoping gives  $(\mathbf{I} - \mathbf{A})(\mathbf{I} + \mathbf{A} + \cdots + \mathbf{A}^K) = \mathbf{I} - \mathbf{A}^{K+1}$ , an exact identity for every  $K$ . If  $\mathbf{I} - \mathbf{A}$  is invertible, solve it for the partial sum and let  $K \rightarrow \infty$ , where [Theorem 10.10.2](#) kills  $\mathbf{A}^{K+1}$ . Conversely, the terms of a convergent series tend to 0, which is the hypothesis of that theorem again.

*Proof.* ( $\Leftarrow$ ) Suppose  $|\lambda| < 1$  for every  $\lambda \in \text{spec}(\mathbf{A})$ . In particular  $1 \notin \text{spec}(\mathbf{A})$ , so  $0 \notin \text{spec}(\mathbf{I} - \mathbf{A})$  — a vector with  $(\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$  is a vector with  $\mathbf{A}\mathbf{v} = \mathbf{v}$  — and  $\mathbf{I} - \mathbf{A}$  is invertible (Theorem 9.2.6). Put  $\mathbf{S}_K = \sum_{k=0}^K \mathbf{A}^k$ . Multiplying out and canceling,

$$(\mathbf{I} - \mathbf{A})\mathbf{S}_K = \sum_{k=0}^K \mathbf{A}^k - \sum_{k=1}^{K+1} \mathbf{A}^k = \mathbf{I} - \mathbf{A}^{K+1},$$

hence  $\mathbf{S}_K = (\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} - \mathbf{A}^{K+1})$ . By Theorem 10.10.2,  $\mathbf{A}^{K+1} \rightarrow 0$ . Each entry of  $\mathbf{S}_K$  is a fixed linear combination of the entries of  $\mathbf{I} - \mathbf{A}^{K+1}$ , with coefficients from  $(\mathbf{I} - \mathbf{A})^{-1}$ , so by (B3) it converges to the corresponding combination for  $\mathbf{I}$ , that is,  $\mathbf{S}_K \rightarrow (\mathbf{I} - \mathbf{A})^{-1}\mathbf{I} = (\mathbf{I} - \mathbf{A})^{-1}$ .

( $\Rightarrow$ ) Suppose  $\sum_k \mathbf{A}^k$  converges. Fix a position  $(r, s)$ ; the series of complex numbers  $\sum_k (\mathbf{A}^k)_{rs}$  converges, so  $(\mathbf{A}^k)_{rs} \rightarrow 0$  by (B4). This holds at every position, so  $\mathbf{A}^k \rightarrow 0$ , and Theorem 10.10.2 gives  $|\lambda| < 1$  for every eigenvalue. This proves the theorem. ■

### Example. Inverting by Summing a Series

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$ . Verify that  $\sum_k \mathbf{A}^k$  converges, sum it, and compare with  $(\mathbf{I} - \mathbf{A})^{-1}$ .

*Solution.*  $p_{\mathbf{A}} = x^2 - \frac{1}{4}$ , so  $\text{spec}(\mathbf{A}) = \{\frac{1}{2}, -\frac{1}{2}\}$  and both eigenvalues have modulus  $\frac{1}{2} < 1$ . By Theorem 10.10.6 the series converges.

*Summing it.*  $\mathbf{A}^2 = \frac{1}{4}\mathbf{I}$ , so  $\mathbf{A}^{2l} = 4^{-l}\mathbf{I}$  and  $\mathbf{A}^{2l+1} = 4^{-l}\mathbf{A}$ . Grouping the partial sums by parity, which is legitimate because each entry sequence splits into two convergent pieces (B3),

$$\sum_{k \geq 0} \mathbf{A}^k = \left( \sum_{l \geq 0} 4^{-l} \right) (\mathbf{I} + \mathbf{A}) = \frac{1}{1 - \frac{1}{4}} (\mathbf{I} + \mathbf{A}) = \frac{4}{3} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix} = \begin{pmatrix} 4/3 & 2/3 \\ 2/3 & 4/3 \end{pmatrix},$$

the scalar sum being the case  $n = 1$  of Theorem 10.10.6 applied to the  $1 \times 1$  matrix  $(\frac{1}{4})$ .

*The inverse.*  $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$  has determinant  $1 - \frac{1}{4} = \frac{3}{4}$ , so by Theorem 0.9.8

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{4}{3} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix},$$

in agreement.

*Remark.* The first thing every criterion in this section looks at is the largest modulus of an eigenvalue,

$$\rho(\mathbf{A}) = \max\{|\lambda| : \lambda \in \text{spec}(\mathbf{A})\},$$

called the **spectral radius** of  $\mathbf{A}$ . Two of the criteria need nothing else:  $\mathbf{A}^m \rightarrow 0$  if and only if  $\rho(\mathbf{A}) < 1$ , and  $\sum_k \mathbf{A}^k$  converges if and only if  $\rho(\mathbf{A}) < 1$ . The other two need more when  $\rho(\mathbf{A}) = 1$ : boundedness also asks for the block sizes at modulus 1, and convergence asks

in addition that 1 be the only eigenvalue of modulus 1. Chapter 16 measures matrices by norms, proves that the spectral radius never exceeds the norm of  $\mathbf{A}$  induced by any vector norm, and recovers  $\rho(\mathbf{A})$  as a limit built from the norms of the powers  $\mathbf{A}^m$ ; that is what turns these exact criteria into usable estimates.

## 10.10.7 Exercises

### 10.10.7.1 A. Check your understanding

#### Exercise. A1

- Define entrywise convergence and entrywise boundedness for a sequence of matrices.
- Write down the entries of  $\mathbf{J}_k(\lambda)^m$ , and say which one shows that a block of size  $\geq 2$  with  $|\lambda| = 1$  has unbounded powers.
- State the three criteria:  $\mathbf{A}^m \rightarrow 0$ ,  $(\mathbf{A}^m)$  bounded,  $(\mathbf{A}^m)$  convergent.
- True or false: if every entry of  $\mathbf{A}$  has modulus less than 1, then  $\mathbf{A}^m \rightarrow 0$ . Justify your answer.
- True or false: if  $(\mathbf{A}^m)$  is bounded then it converges. Justify your answer.
- For which  $\mathbf{A}$  does  $\sum_k \mathbf{A}^k$  converge, and what is the sum?

*Solution.*

- $\mathbf{M}_m \rightarrow \mathbf{M}$  if  $(\mathbf{M}_m)_{rs} \rightarrow \mathbf{M}_{rs}$  for every position  $(r, s)$ ;  $(\mathbf{M}_m)$  is bounded if there is one constant  $c$  with  $|(\mathbf{M}_m)_{rs}| \leq c$  for all  $m$  and all  $(r, s)$  (Definition 10.10.1).
- $(\mathbf{J}_k(\lambda)^m)_{r, r+j} = \binom{m}{j} \lambda^{m-j}$  for  $0 \leq j \leq k-1$ , with zeros below the diagonal (Lemma 10.9.1 with  $p = x^m$ ). The entry with  $j = 1$  is  $m\lambda^{m-1}$ , of modulus  $m$  when  $|\lambda| = 1$ .
- $\mathbf{A}^m \rightarrow 0$  iff every eigenvalue has  $|\lambda| < 1$  (Theorem 10.10.2);  $(\mathbf{A}^m)$  is bounded iff every eigenvalue has  $|\lambda| \leq 1$  and those with  $|\lambda| = 1$  have only  $1 \times 1$  blocks (Theorem 10.10.3);  $(\mathbf{A}^m)$  converges iff every eigenvalue has  $|\lambda| < 1$  or equals 1 with only  $1 \times 1$  blocks (Theorem 10.10.4).
- False.  $\mathbf{A} = \begin{pmatrix} 0.9 & 0.9 \\ 0.9 & 0.9 \end{pmatrix}$  has the eigenvalue 1.8, and  $\mathbf{A}^m \mathbf{1} = 1.8^m \mathbf{1}$  is unbounded; see the warning after Theorem 10.10.2.
- False.  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies  $\mathbf{A}^2 = \mathbf{I}$ , so  $\mathbf{A}^m$  alternates between  $\mathbf{I}$  and  $\mathbf{A}$  and is bounded but not convergent. Its eigenvalues are  $\pm 1$ , and the eigenvalue  $-1$  is what Theorem 10.10.4 excludes.
- Exactly when every eigenvalue has  $|\lambda| < 1$ , and then the sum is  $(\mathbf{I} - \mathbf{A})^{-1}$  (Theorem 10.10.6).

### 10.10.7.2 B. Practice

#### Exercise. B1: Decide the behavior

For each matrix over  $\mathbb{C}$ , decide whether  $\mathbf{A}^m \rightarrow 0$ , whether  $(\mathbf{A}^m)$  is bounded, and whether

$(\mathbf{A}^m)$  converges. Justify your answer.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \mathbf{J}_2(1) \oplus \mathbf{J}_1(\frac{1}{2}) \in M_3(\mathbb{C})$ .
- (d)  $\mathbf{A}_4 = \begin{pmatrix} 0 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a) The only eigenvalue is  $\frac{1}{2}$ , of modulus  $< 1$ , so  $\mathbf{A}_1^m \rightarrow 0$  by [Theorem 10.10.2](#); hence it is bounded and convergent. (The superdiagonal 1 is irrelevant: by Lemma 10.9.1 the  $(1, 2)$ -entry of  $\mathbf{A}_1^m$  is  $m 2^{-(m-1)} \rightarrow 0$  by (B1).)
- (b)  $p_{\mathbf{A}_2} = x^2 + 1$ , so  $\text{spec}(\mathbf{A}_2) = \{i, -i\}$ , both of modulus 1, and the eigenvalues are distinct, so every block has size 1. By [Theorem 10.10.3](#),  $(\mathbf{A}_2^m)$  is **bounded**; it does not tend to 0 ([Theorem 10.10.2](#)) and does not converge, since the eigenvalue  $i$  has modulus 1 and is not 1 ([Theorem 10.10.4](#)). Concretely  $\mathbf{A}_2^4 = \mathbf{I}$ , so the powers cycle through four matrices.
- (c) The eigenvalue 1 has a block of size 2, so [Theorem 10.10.3](#) (ii) fails:  $(\mathbf{A}_3^m)$  is **unbounded**, hence neither convergent nor tending to 0. The  $(1, 2)$ -entry of  $\mathbf{A}_3^m$  is  $m$ .
- (d)  $\mathbf{A}_4^2 = 0$ , so  $\mathbf{A}_4^m = 0$  for  $m \geq 2$  and  $\mathbf{A}_4^m \rightarrow 0$  trivially: all eigenvalues are 0. The large entry 5 does not matter, illustrating the warning after [Theorem 10.10.2](#).

### Exercise. B2: Computing limits

- (a) Let  $\mathbf{A} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ . Show that  $\mathbf{A}^m$  converges and find the limit, both by [Theorem 10.10.4](#) and by a direct computation of  $\mathbf{A}^m$ .
- (b) Let  $\mathbf{B} = \begin{pmatrix} 1 & 1/3 \\ 0 & 1/3 \end{pmatrix}$ . Find  $\lim_m \mathbf{B}^m$  and identify it as a projection.

*Solution.*

- (a)  $p_{\mathbf{A}} = x^2 - x = x(x - 1)$ , so  $\text{spec}(\mathbf{A}) = \{0, 1\}$ . The eigenvalue 0 has  $|0| < 1$  and the eigenvalue 1 has  $a_{\mathbf{A}}(1) = 1$ , so its single block has size 1; by [Theorem 10.10.4](#) the powers converge, and the limit is the projection onto  $E_1(\mathbf{A})$  along  $G_0(\mathbf{A}) = E_0(\mathbf{A})$ . Now  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$  gives  $E_1(\mathbf{A}) = \text{span}((1, 1))$  and  $E_0(\mathbf{A}) = \ker \mathbf{A} =$

$\text{span}((1, -1))$ . With  $\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{Q}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,

$$\lim_m \mathbf{A}^m = \mathbf{Q} \text{diag}(1, 0) \mathbf{Q}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{A}.$$

*Directly.*  $\mathbf{A}^2 = \mathbf{A}$ , so  $\mathbf{A}^m = \mathbf{A}$  for every  $m \geq 1$  and the limit is  $\mathbf{A}$ . ( $\mathbf{A}$  is itself a projection, and the theorem returns it unchanged.)

- (b)  $\mathbf{B}$  is upper triangular, so  $\text{spec}(\mathbf{B}) = \{1, \frac{1}{3}\}$  (Theorem 7.4.6). Both criteria hold:  $|\frac{1}{3}| < 1$ , and  $a_{\mathbf{B}}(1) = 1$  so the 1-block has size 1. Solving  $(\mathbf{B} - \mathbf{I})\mathbf{x} = \mathbf{0}$  with  $\mathbf{B} - \mathbf{I} = \begin{pmatrix} 0 & 1/3 \\ 0 & -2/3 \end{pmatrix}$  gives  $E_1(\mathbf{B}) = \text{span}(\mathbf{e}_1)$ ; solving  $(\mathbf{B} - \frac{1}{3}\mathbf{I})\mathbf{x} = \mathbf{0}$  with  $\mathbf{B} - \frac{1}{3}\mathbf{I} = \begin{pmatrix} 2/3 & 1/3 \\ 0 & 0 \end{pmatrix}$  gives  $E_{1/3}(\mathbf{B}) = \text{span}((1, -2))$ . With  $\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}$  we have  $\det \mathbf{Q} = -2$  and  $\mathbf{Q}^{-1} = \frac{1}{-2} \begin{pmatrix} -2 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1/2 \\ 0 & -1/2 \end{pmatrix}$ . Multiplying in two steps,  $\mathbf{Q} \text{diag}(1, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and then

$$\lim_m \mathbf{B}^m = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{Q}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix}.$$

This is the projection onto  $\text{span}(\mathbf{e}_1)$  along  $\text{span}((1, -2))$ : it fixes  $\mathbf{e}_1$  and sends  $(1, -2)$  to  $(1 - 1, 0) = \mathbf{0}$ . *Check.*  $\mathbf{B}^m = \begin{pmatrix} 1 & c_m \\ 0 & 3^{-m} \end{pmatrix}$  with  $c_1 = \frac{1}{3}$  and  $c_{m+1} = \frac{1}{3}(1 + c_m)$ , read off  $\mathbf{B}^{m+1} = \mathbf{B}^m \mathbf{B}$ ; the first values are  $\frac{1}{3}, \frac{4}{9}, \frac{13}{27}$ , climbing towards the fixed point  $c = \frac{1}{3}(1 + c)$ , that is  $c = \frac{1}{2}$ .

### Exercise. B3: A Neumann series

Let  $\mathbf{A} = \begin{pmatrix} 1/3 & 1/3 \\ 0 & 1/3 \end{pmatrix}$ .

- Explain why  $\sum_k \mathbf{A}^k$  converges.
- Compute  $(\mathbf{I} - \mathbf{A})^{-1}$  directly.
- Compute the partial sum  $\mathbf{I} + \mathbf{A} + \mathbf{A}^2$  and compare it with (b).

*Solution.*

- $\mathbf{A}$  is upper triangular with both diagonal entries  $\frac{1}{3}$ , so  $\text{spec}(\mathbf{A}) = \{\frac{1}{3}\}$  (Theorem 7.4.6) and  $|\frac{1}{3}| < 1$ . By Theorem 10.10.6 the series converges, with sum  $(\mathbf{I} - \mathbf{A})^{-1}$ .
- $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 2/3 & -1/3 \\ 0 & 2/3 \end{pmatrix}$ , of determinant  $\frac{4}{9}$ . By Theorem 0.9.8,

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{9}{4} \begin{pmatrix} 2/3 & 1/3 \\ 0 & 2/3 \end{pmatrix} = \begin{pmatrix} 3/2 & 3/4 \\ 0 & 3/2 \end{pmatrix}.$$

- (c) Write  $\mathbf{A} = \frac{1}{3}(\mathbf{I} + \mathbf{M})$  with  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{M}^2 = 0$ ; then  $\mathbf{A}^k = 3^{-k}(\mathbf{I} + k\mathbf{M})$  by the binomial theorem, the two summands commuting. So

$$\mathbf{I} + \mathbf{A} + \mathbf{A}^2 = (1 + \frac{1}{3} + \frac{1}{9})\mathbf{I} + (\frac{1}{3} + \frac{2}{9})\mathbf{M} = \frac{13}{9}\mathbf{I} + \frac{5}{9}\mathbf{M} = \begin{pmatrix} 13/9 & 5/9 \\ 0 & 13/9 \end{pmatrix}.$$

Summing the whole series the same way,  $(\mathbf{I} - \mathbf{A})^{-1} = (\sum_{k \geq 0} 3^{-k})\mathbf{I} + (\sum_{k \geq 0} k3^{-k})\mathbf{M}$ , so comparing with (b) reads off  $\sum_k 3^{-k} = \frac{3}{2}$  and  $\sum_k k3^{-k} = \frac{3}{4}$ . The two partial sums  $\frac{13}{9}$  and  $\frac{5}{9}$ , about 1.44 and 0.56, are on their way to  $\frac{3}{2}$  and  $\frac{3}{4}$ ; the convergence is geometric, as the factor  $3^{-k}$  shows.

### 10.10.7.3 C. Going deeper

#### Exercise. C1: Bounded but not convergent

- Give an example of  $\mathbf{A} \in M_2(\mathbb{R})$  with  $(\mathbf{A}^m)$  bounded but not convergent, and one with  $(\mathbf{A}^m)$  convergent but  $\mathbf{A}^m \not\rightarrow 0$ .
- Prove that if  $\mathbf{A}^m \rightarrow \mathbb{L}$ , then  $\mathbb{L}^2 = \mathbb{L}$  and  $\mathbf{A}\mathbb{L} = \mathbb{L}\mathbf{A} = \mathbb{L}$ .
- Deduce that if  $\mathbf{A}^m \rightarrow \mathbb{L}$  with  $\mathbb{L}$  **invertible**, then  $\mathbb{L} = \mathbf{I}$  and  $\mathbf{A} = \mathbf{I}$ .
- Give an invertible  $\mathbf{A}$  with  $\mathbf{A}^m \rightarrow \mathbb{L}$  and  $\mathbb{L} \notin \{0, \mathbf{I}\}$ , so that the invertibility of  $\mathbb{L}$ , not of  $\mathbf{A}$ , is what part (c) needs.

*Solution.*

- For the first,  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ :  $\mathbf{A}^2 = \mathbf{I}$ , so  $\mathbf{A}^m$  alternates between  $\mathbf{I}$  and  $\mathbf{A}$  and is bounded but not convergent; its eigenvalues are  $\pm 1$ . For the second,  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$ :  $\mathbf{A}^m = \text{diag}(1, 2^{-m}) \rightarrow \text{diag}(1, 0) \neq 0$ .
- Each entry of  $\mathbf{A}^m \mathbf{A}^m = \mathbf{A}^{2m}$  is a sum of products of entries of  $\mathbf{A}^m$ , so by (B3) it converges to the corresponding expression in the entries of  $\mathbb{L}$ ; hence  $\mathbf{A}^{2m} \rightarrow \mathbb{L}^2$ . But  $(\mathbf{A}^{2m})_m$  is a subsequence of  $(\mathbf{A}^m)_m$  — more precisely, for each position the sequence  $(\mathbf{A}^{2m})_{rs}$  is a subsequence of the convergent sequence  $(\mathbf{A}^m)_{rs}$ , so it has the same limit  $\mathbb{L}_{rs}$ . Hence  $\mathbb{L}^2 = \mathbb{L}$ . Similarly  $\mathbf{A}^{m+1} = \mathbf{A}\mathbf{A}^m = \mathbf{A}^m \mathbf{A}$ , and the left side is a tail of the convergent sequence, so it tends to  $\mathbb{L}$ , while the two right sides tend to  $\mathbf{A}\mathbb{L}$  and  $\mathbb{L}\mathbf{A}$  by (B3). Hence  $\mathbf{A}\mathbb{L} = \mathbb{L}\mathbf{A} = \mathbb{L}$ .
- Suppose  $\mathbf{A}^m \rightarrow \mathbb{L}$  with  $\mathbb{L}$  invertible. By (b),  $\mathbb{L}^2 = \mathbb{L}$ ; multiplying by  $\mathbb{L}^{-1}$  gives  $\mathbb{L} = \mathbf{I}$ . Then  $\mathbf{A}\mathbb{L} = \mathbb{L}$  reads  $\mathbf{A} = \mathbf{I}$ . (Conversely  $\mathbf{A} = \mathbf{I}$  gives  $\mathbf{A}^m = \mathbf{I} \rightarrow \mathbf{I}$ , so this case really occurs.)
- $\mathbf{A} = \text{diag}(1, \frac{1}{2})$  is invertible, and  $\mathbf{A}^m = \text{diag}(1, 2^{-m}) \rightarrow \text{diag}(1, 0) = \mathbb{L}$ , which is neither 0 nor  $\mathbf{I}$  — and not invertible. So no hypothesis on  $\mathbf{A}$  alone forces  $\mathbb{L} \in \{0, \mathbf{I}\}$ : by [Theorem 10.10.4](#) the limit is the projection onto  $E_1(\mathbf{A})$  along the other generalized eigenspaces, and that projection is 0 or  $\mathbf{I}$  only in the extreme cases  $1 \notin \text{spec}(\mathbf{A})$  and  $E_1(\mathbf{A}) = \mathbb{C}^n$ .

**Exercise. C2: Chapter 9's Markov theorem without diagonalizability**

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic with  $a_{\mathbf{A}}(1) = 1$  and  $|\lambda| < 1$  for every eigenvalue  $\lambda \neq 1$ .

- (a) Prove that  $\mathbf{1}^\top \mathbf{w} = 0$  for every  $\mathbf{w} \in G_\lambda(\mathbf{A})$  with  $\lambda \neq 1$ . *Hint:*  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda) \mathbf{1}^\top$ .
- (b) Prove that  $(\mathbf{A}^m)$  converges to a **stochastic** matrix  $\mathbf{\Pi}$ , and deduce that  $\mathbf{A}$  has a steady state  $\mathbf{v}$  with  $E_1(\mathbf{A}) = \text{span}(\mathbf{v})$ .
- (c) Deduce that  $\mathbf{\Pi} = \mathbf{v} \mathbf{1}^\top$  and that  $\mathbf{A}^m \mathbf{x}_0 \rightarrow \mathbf{v}$  for every probability vector  $\mathbf{x}_0$ , **without** assuming that  $\mathbf{A}$  is diagonalizable.

*Solution.*

- (a) Since  $\mathbf{A}$  is stochastic,  $\mathbf{1}^\top \mathbf{A} = \mathbf{1}^\top$  (Proposition 9.11.3 (c)), so  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I}) = \mathbf{1}^\top - \lambda \mathbf{1}^\top = (1 - \lambda) \mathbf{1}^\top$ . Iterating  $n$  times,  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I})^n = (1 - \lambda)^n \mathbf{1}^\top$ . Let  $\mathbf{w} \in G_\lambda(\mathbf{A})$ , so  $(\mathbf{A} - \lambda \mathbf{I})^n \mathbf{w} = \mathbf{0}$  (Corollary 10.1.2 (a)). Then

$$0 = \mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I})^n \mathbf{w} = (1 - \lambda)^n \mathbf{1}^\top \mathbf{w},$$

and  $\lambda \neq 1$  gives  $(1 - \lambda)^n \neq 0$ , so  $\mathbf{1}^\top \mathbf{w} = 0$ .

- (b) By hypothesis 1 is the only eigenvalue of modulus 1, so  $(\mathbf{A}^m)$  converges by [Corollary 10.10.5](#) (b); call the limit  $\mathbf{\Pi}$ . Every  $\mathbf{A}^m$  is stochastic (Proposition 9.11.3 (b)), so all its entries are  $\geq 0$  and each of its columns sums to 1. Both properties pass to the limit: a limit of real numbers  $\geq 0$  is  $\geq 0$ , and a column sum is a finite sum of entries, so it converges to the sum of the limits (Chapter 9 §11). Hence  $\mathbf{\Pi}$  is stochastic, and in particular  $\mathbf{\Pi} \neq \mathbf{0}$ .

By [Theorem 10.10.4](#),  $\mathbf{\Pi}$  is the projection onto  $E_1(\mathbf{A})$  along  $\bigoplus_{\lambda \neq 1} G_\lambda(\mathbf{A})$ , so  $\text{im } T_{\mathbf{\Pi}} = E_1(\mathbf{A})$ . Since  $a_{\mathbf{A}}(1) = 1$  and  $1 \leq g_{\mathbf{A}}(1) \leq a_{\mathbf{A}}(1)$  (Theorem 9.3.1),  $E_1(\mathbf{A})$  is a line. Each column of  $\mathbf{\Pi}$  lies in  $\text{im } T_{\mathbf{\Pi}} = E_1(\mathbf{A})$  (Definition 0.9.9: column  $j$  is  $\mathbf{\Pi} \mathbf{e}_j$ ) and is a probability vector, hence is fixed by  $\mathbf{A}$  and is a steady state. Any two probability vectors on the same line are equal: if  $\mathbf{w} = c\mathbf{v}$  with both entry sums equal to 1, then  $1 = c$ . So all the columns of  $\mathbf{\Pi}$  are one and the same steady state  $\mathbf{v}$ , and  $E_1(\mathbf{A}) = \text{span}(\mathbf{v})$ .

- (c) All columns of  $\mathbf{\Pi}$  equal  $\mathbf{v}$ , which is exactly the statement  $\mathbf{\Pi} = \mathbf{v} \mathbf{1}^\top$ . (Independently of (b), part (a) confirms it:  $\mathbf{v} \mathbf{1}^\top$  fixes  $\mathbf{v}$  because  $\mathbf{1}^\top \mathbf{v} = 1$ , and kills every  $\mathbf{w} \in G_\lambda(\mathbf{A})$  with  $\lambda \neq 1$  because  $\mathbf{1}^\top \mathbf{w} = 0$ ; by Theorem 10.1.3 (a) those two conditions determine the projection onto  $E_1(\mathbf{A}) = G_1(\mathbf{A})$  along the rest, using  $\dim G_1(\mathbf{A}) = a_{\mathbf{A}}(1) = 1 = \dim E_1(\mathbf{A})$  and Theorem 1.6.8.) For a probability vector  $\mathbf{x}_0$ , each entry of  $\mathbf{A}^m \mathbf{x}_0$  is a fixed linear combination of entries of  $\mathbf{A}^m$ , so by (B3)

$$\mathbf{A}^m \mathbf{x}_0 \longrightarrow \mathbf{\Pi} \mathbf{x}_0 = \mathbf{v} (\mathbf{1}^\top \mathbf{x}_0) = \mathbf{v} (x_{01} + \cdots + x_{0n}) = \mathbf{v}.$$

No diagonalizability was used: the blocks for the eigenvalues  $\lambda \neq 1$  may have any sizes, since [Theorem 10.10.2](#) does not care. This removes the diagonalizability hypothesis from Theorem 9.11.4.